

Bare Advanced Demo of IEEEtran.cls for IEEE Computer Society Journals

Michael Shell, *Member, IEEE*, John Doe, *Fellow, OSA*, and Jane Doe, *Life Fellow, IEEE*

Abstract—The abstract goes here.

Index Terms—Computer Society, IEEE, IEEEtran, journal, L^AT_EX, paper, template.

1 INTRODUCTION

MULTI-LABEL image tagging is a practical recognition problem in which an image must be assigned several semantic concepts rather than one exclusive class. This setting is common in web image search, media organization, and content recommendation, where a single image may contain objects, scenes, textures, and contextual attributes at the same time. Unlike single-label classification, image tagging must decide both which labels are present and how multiple labels interact. This makes the task sensitive to class imbalance, weak visual evidence for rare concepts, and correlations among labels [1], [2], [3].

The 81-label image tagging task studied in this project reflects these challenges. Each image is represented by heterogeneous handcrafted visual features: a local Bag-of-Words (BoW) descriptor and a group of global color-texture descriptors. These features encode different visual cues, but their relative usefulness is not obvious before experimentation. In addition, each image is associated with only a small subset of the 81 possible tags, so the model observes many more negative label decisions than positive ones. A standard binary cross-entropy objective can therefore be dominated by easy negative labels, while a visually plausible prediction may still fail on infrequent tags.

Prior work suggests three complementary directions for this problem. First, imbalance-aware losses such as Focal Loss and Asymmetric Loss modify the training objective so that hard or positive examples receive more useful gradient signal [4], [5]. Second, feature fusion can combine local and global image descriptors before prediction, allowing the classifier to use complementary evidence from different visual representations [6], [7], [8]. Third, label-dependency methods exploit the fact that tags are not independent; some labels frequently co-occur because they describe related objects, scenes, or contexts [9], [10]. These ideas motivate a controlled experimental study rather than a single model choice.

This report investigates multi-label image tagging through two design axes: the visual representation and the loss function. On the representation side, we compare a

BoW-only baseline, a Color+Texture baseline, simple early fusion, gated fusion, gated fusion with an InfoNCE alignment term, and an additional co-occurrence-aware ASL model. On the optimization side, we compare binary cross-entropy, Focal Loss, and Asymmetric Loss under the same 81-label evaluation setting. Performance is measured using mean average precision (mAP), Micro-F1, and Macro-F1, so that ranking quality, overall label classification, and rare-label behavior are all considered.

The experiments show several consistent patterns. Color+Texture features are substantially stronger than BoW features as a standalone representation. Simple feature fusion improves or matches the strongest single-feature baseline, especially for mAP, while the InfoNCE alignment term hurts performance across losses, suggesting that forcing heterogeneous handcrafted features into a shared contrastive space can discard useful modality-specific information. Among loss functions, Focal Loss produces the best mAP (36.67%) with early fusion, whereas Asymmetric Loss gives the best Micro-F1 (63.47%) and Macro-F1 (33.98%), indicating that it is better suited to the imbalanced multi-label classification objective. These findings support the central argument of the report: effective image tagging requires considering feature fusion, imbalance-aware optimization, and label dependency together rather than treating them as isolated components.

2 RELATED WORK

Multi-label image tagging assigns a set of semantic labels to each image, rather than forcing the image into a single mutually exclusive class. Early multi-label learning work established this setting as a structured prediction problem in which label relevance, ranking, and dependency all matter [1], [2]. Modern image tagging benchmarks such as PASCAL VOC, NUS-WIDE, MS-COCO, VG-500, and News-500 further show that the problem becomes more difficult as the label space grows from tens of object categories to hundreds of object, scene, action, and abstract concepts [3], [11], [12], [13], [14]. In large label spaces, image tags are not independent: objects co-occur with scenes, actions, and contextual concepts. As a result, recent work has increasingly moved beyond independent binary classifiers toward models that

• M. Shell was with the Department of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, 30332.
E-mail: see <http://www.michaelsshell.org/contact.html>
• J. Doe and J. Doe are with Anonymous University.
Manuscript received April 19, 2005; revised August 26, 2015.

account for label correlation, long-tailed distributions, and heterogeneous visual representations.

2.1 Imbalance-Aware Multi-Label Learning

Class imbalance is a central difficulty in multi-label image tagging. Each image usually contains only a small number of positive labels and many negative labels, and web-scale datasets often follow a long-tailed distribution in which frequent categories dominate optimization while rare categories receive weak supervision [15], [16]. Data-level strategies such as over-sampling, under-sampling, and synthetic sample generation attempt to rebalance the training distribution before learning [15]. Although these methods are simple and useful in conventional classification, they are less straightforward in multi-label image tagging because labels are semantically coupled: duplicating or removing an image changes the evidence for several labels at once and may distort co-occurrence structure.

Algorithm-level methods instead modify the training objective. Binary cross-entropy is a common baseline, but in imbalanced multi-label settings it is often dominated by abundant easy negatives. Focal-style losses reduce this dominance by emphasizing harder examples, while Asymmetric Loss further separates the treatment of positive and negative labels for multi-label recognition [4], [5]. Distribution-Balanced Loss explicitly targets long-tailed multi-label learning by correcting both label-frequency imbalance and negative-tolerant regularization [17]. These losses improve optimization, but they do not fully solve rare-label recognition when the underlying visual features for minority concepts are weak or when rare labels depend on semantic context from other labels [18]. Thus, imbalance handling is most effective when it is considered together with representation learning and label dependency modeling.

2.2 Label Co-Occurrence and Semantic Dependency

Another major line of work models label relationships. Second-order approaches capture pairwise dependencies and are attractive because they balance expressiveness and computational cost. Rank-SVM introduced a ranking formulation for multi-labelled classification, encouraging relevant labels to score above irrelevant labels [2]. Later pairwise ranking methods adapted this idea to deep multi-label image classification and improved threshold estimation for prediction [19]. Other approaches inject empirical co-occurrence statistics into neural models, either as priors or as post-processing calibration, so that predictions become more consistent with observed label relations [10], [20], [21].

Pairwise statistics, however, cannot represent higher-order contexts where a label is meaningful only through a combination of several other labels. Classifier Chains model this dependency by predicting labels sequentially and conditioning later classifiers on earlier outputs [22]. Probabilistic Classifier Chains extend this idea by modeling the joint label distribution more explicitly [23]. In deep image recognition, graph and attention-based methods provide more flexible high-order dependency modeling. ML-GCN represents labels as graph nodes and propagates information through graph convolution [9], while transformer-based

models such as C-Tran use learnable label embeddings and self-attention to capture global label interactions [3]. These methods improve semantic reasoning, but they can inherit bias from imbalanced training data: co-occurrence estimates are usually strongest for frequent labels, so tail-label relations may remain underrepresented.

2.3 Feature Fusion for Multi-Label Tagging

Feature fusion addresses the representation side of multi-label tagging. A single visual feature source may be insufficient for complex scenes, where object appearance, texture, spatial layout, and contextual cues jointly determine the label set. General surveys on multimodal learning and data fusion distinguish early fusion, which combines raw or low-level features before high-level abstraction, from deeper fusion strategies that align learned representations through attention, graph, or transformer modules [6], [7], [8], [24]. Early fusion can exploit complementary cues from the beginning of the learning process, and recent mixed-modal or cross-aware transformer designs extend this principle to tokenized and intermediate representations [25], [26], [27].

Deep fusion methods operate in learned feature spaces and are better suited to semantic alignment across heterogeneous sources. Attention mechanisms can adaptively weight features relevant to specific tags, graph structures can propagate relational information, and transformer modules can model global interactions among visual regions and label embeddings [24], [28]. Nevertheless, fusion alone does not guarantee robust multi-label prediction under long-tailed distributions. Large fusion modules may overfit when high-quality labels are scarce, and frequent-label visual patterns can dominate joint representations. Therefore, current literature suggests that fusion, imbalance-aware learning, and semantic dependency modeling should be integrated rather than treated as independent components.

2.4 Research Gap

The literature survey indicates a common limitation across these directions: most existing methods address only one aspect of the problem. Loss-based imbalance methods improve optimization but may ignore label semantics; co-occurrence and graph-based models improve dependency reasoning but can amplify head-label bias; fusion models enrich representations but may still underrepresent rare concepts. This motivates a unified framework for our project, where multi-source visual features are fused with label-dependency modeling and trained using imbalance-aware objectives. Such a design is better aligned with large-scale real-world image tagging, where rare labels, semantic correlations, and heterogeneous visual evidence interact within the same prediction problem.

3 METHODOLOGY

4 EXPERIMENT RESULT

5 DISCUSSION

6 FURTHER DIRECTION

final

7 CONCLUSION

The conclusion goes here.

APPENDIX A

PROOF OF THE FIRST ZONKLAR EQUATION

Appendix one text goes here.

APPENDIX B

Appendix two text goes here.

ACKNOWLEDGMENTS

The authors would like to thank...

REFERENCES

- [1] M.-L. Zhang and Z.-H. Zhou, "A k-nearest neighbor based algorithm for multi-label classification," in *2005 IEEE International Conference on Granular Computing*, vol. 2. IEEE, 2005, pp. 718–721.
- [2] A. Elisseeff and J. Weston, "A kernel method for multi-labelled classification," in *Advances in Neural Information Processing Systems*, vol. 14, 2001.
- [3] J. Lanchantin, T. Wang, V. Ordonez, and Y. Qi, "General multi-label image classification with transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 16 478–16 488.
- [4] Q. Wang, Y. Ma, K. Zhao, and Y. Tian, "A comprehensive survey of loss functions in machine learning," *Annals of Data Science*, vol. 9, no. 2, pp. 187–212, 2022.
- [5] T. Ridnik, E. Ben-Baruch, N. Zamir, A. Noy, I. Friedman, M. Protter, and L. Zelnik-Manor, "Asymmetric loss for multi-label classification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 82–91.
- [6] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2018.
- [7] T. Meng, X. Jing, Z. Yan, and W. Pedrycz, "A survey on machine learning for data fusion," *Information Fusion*, vol. 57, pp. 115–129, 2020.
- [8] J. Gao, P. Li, Z. Chen, and J. Zhang, "A survey on deep learning for multimodal data fusion," *Neural Computation*, vol. 32, no. 5, pp. 829–864, 2020.
- [9] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5177–5186.
- [10] S. Rawlekar, S. Bhatnagar, V. P. Srinivasulu, and N. Ahuja, "Improving multi-label recognition using class co-occurrence probabilities," in *International Conference on Pattern Recognition*. Springer, 2024, pp. 424–439.
- [11] M. Everingham and J. Winn, "The pascal visual object classes challenge 2012 (voc2012) development kit," *Pattern Analysis, Statistical Modelling and Computational Learning*, Tech. Rep., 2011.
- [12] T.-S. Chua, J. Tang, R. Hong, H. Li, Z. Luo, and Y.-T. Zheng, "Nus-wide: A real-world web image database from national university of singapore," in *Proceedings of the ACM International Conference on Image and Video Retrieval*, 2009, pp. 1–9.
- [13] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European Conference on Computer Vision*. Springer, 2014, pp. 740–755.
- [14] T. Chen, L. Lin, R. Chen, X. Hui, and H. Wu, "Knowledge-guided multi-label few-shot learning for general image recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 3, pp. 1371–1384, 2020.
- [15] M. Altalhan, A. Algarni, and M. T.-H. Alouane, "Imbalanced data problem in machine learning: A review," *IEEE Access*, vol. 13, pp. 13 686–13 699, 2025.
- [16] B. Li, Z. Han, H. Li, H. Fu, and C. Zhang, "Trustworthy long-tailed classification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 6970–6979.
- [17] T. Wu, Q. Huang, Z. Liu, Y. Wang, and D. Lin, "Distribution-balanced loss for multi-label classification in long-tailed datasets," in *European Conference on Computer Vision*. Springer, 2020, pp. 162–178.
- [18] T. Li, P. Cao, Y. Yuan, L. Fan, Y. Yang, R. S. Feris, P. Indyk, and D. Katabi, "Targeted supervised contrastive learning for long-tailed recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 6918–6928.
- [19] Y. Li, Y. Song, and J. Luo, "Improving pairwise ranking for multi-label image classification," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 3617–3625.
- [20] J. Yao, K. Wang, and J. Yan, "Incorporating label co-occurrence into neural network-based models for multi-label text classification," *IEEE Access*, vol. 7, pp. 183 580–183 588, 2019.
- [21] X. Zhu, J. Liu, W. Liu, J. Ge, B. Liu, and J. Cao, "Scene-aware label graph learning for multi-label image classification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 1473–1482.
- [22] J. Read, B. Pfahringer, G. Holmes, and E. Frank, "Classifier chains for multi-label classification," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2009, pp. 254–269.
- [23] W. Cheng, E. Hüllermeier, and K. J. Dembczynski, "Bayes optimal multilabel classification via probabilistic classifier chains," in *Proceedings of the 27th International Conference on Machine Learning*, 2010, pp. 279–286.
- [24] W. Guo, J. Wang, and S. Wang, "Deep multimodal representation learning: A survey," *IEEE Access*, vol. 7, pp. 63 373–63 394, 2019.
- [25] Chameleon Team, "Chameleon: Mixed-modal early-fusion foundation models," *arXiv preprint arXiv:2405.09818*, 2024.
- [26] Y. Cho, H. Yu, and S.-J. Kang, "Cross-aware early fusion with stage-divided vision and language transformer encoders for referring image segmentation," *IEEE Transactions on Multimedia*, vol. 26, pp. 5823–5833, 2023.
- [27] X. V. Lin, A. Shrivastava, L. Luo, S. Iyer, M. Lewis, G. Ghosh, L. Zettlemoyer, and A. Aghajanyan, "Moma: Efficient early-fusion pre-training with mixture of modality-aware experts," *arXiv preprint arXiv:2407.21770*, 2024.
- [28] F. Zhao, C. Zhang, and B. Geng, "Deep multimodal data fusion," *ACM Computing Surveys*, vol. 56, no. 9, pp. 1–36, 2024.



Michael Shell Biography text here.

John Doe Biography text here.

Jane Doe Biography text here.